Protein Sequence Analysis and Secondary Structure Prediction Using Deep Learning

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***Abstract*—Predicting secondary structure from protein sequence is one of the fundamental problems in structural bioinformatics. Our approach uses a biLSTM network to predict secondary structure from sequence data, using a three-state classification scheme of helix, strand, and coil. Our dataset consists of 95,000 protein sequences, of which 55,000 had sequence ambiguity and required preprocessing. After five epochs of training, we achieved 81.53% Q3 accuracy on the validation set, significantly better than classical methods using the Chou-Fasman rules (65%) and early attempts using neural networks (76-78%). For functional classification, we used a Multinomial Naive Bayes classifier to predict functional classes from sequence data, achieving 77% overall accuracy for 42 functional classes. Our results indicate that sequence-based deep learning can compete with other methods, even without using evolution, and is a computationally efficient tool for proteome annotation.**

***Index Terms*—Keywords: protein prediction, bidirectional LSTM, sequence mining, deep learning**

I. Introduction

A. Problem Statement

This is due to an exponential increase in genomics and proteomics data. Currently, there are 200,000 entries in structural databases, while there are hundreds of millions of protein sequences in sequence databases [6], [29]. This is a strong argument for the need of a new approach to bridge the gap between protein sequence and structure.

A new approach for predicting secondary protein structures from their amino acid sequences is a feasible substitute for expensive and time-consuming experiments. The analysis of biological sequences is a special problem. The number of elements is small, only 20 standard amino acids, and pattern length can vary from hundreds to thousands of residues. The accuracy of rule-based approaches is low, making it necessary to resort to a new class of learning approaches: deep learning. [7], [15].

B. Background and Motivation

Bioinformatics is a field of study that brings together information technology and biology to store, manage, and analyze large-scale biological information. The increasing size of genome and

proteome information stores will continue to fuel the need for high-performance computing tools [6], [10].

A protein is a chain of one or many amino acids. The standard set of 20 amino acid residues used for protein construction is: A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, W, Y. The function of a protein is closely linked with its three dimensional structure. The three-dimensional structure of a protein is determined by a hierarchy of information contained in its primary sequence of amino acids. The hierarchy is important for understanding biological mechanisms, diagnosing diseases, and developing drugs [1], [9].

Protein structures exist in four levels. In the primary structure, amino acids are connected by peptide bonds. In the secondary structure, local structures like alpha helices, beta strands, and coils are formed. These structures are formed by hydrogen bonds. In the structure of tertiary, the overall structure of the single polypeptide chain is described. In alpha helices, the structure is right-handed and consists of a spiral structure held together by intrachain hydrogen bonds. In the beta structure, the strands are connected laterally and can occur in parallel or antiparallel arrangements. It takes months or years of dedicated work to determine the structure experimentally [29], [24].

II. Related Work

Classical methods of secondary protein structure prediction have undergone through huge changes in the last four decades or so. The Dictionary of Secondary Structure of Protein algorithm, introduced by Kabsch and Sander, set the benchmark for predicting the secondary structure of protein structures determined by experimental methods, taking into account the patterns of hydrogen bonding and geometric considerations. The algorithm defined eight secondary protein structure classes, which are reduced into three in most cases of prediction methods.

Earlier computer-based methods of protein secondary structure prediction were based on statistical features learned from protein structures whose three-dimensional details were already determined. The Chou-Fasman method employed amino acid conformational parameters for predicting protein secondary structures by considering the probability of occurrence of helices, strands, and coils for each type of amino acid residue in a protein sequence, based on their ability to form helices, strands, and coils. The GOR (Garnier-Osguthorpe-Robson) method improved upon the Chou-Fasman method by taking into account the correlation of amino acids in a protein sequence, including sequence windows. These methods, which pioneered protein secondary structure prediction, have a limited accuracy of 65

The advent of machine learning brought a tremendous leap forward in the prediction of secondary structure. Scientists have used classification algorithms like native Bayes, Neural Networks, Genetic Algorithm, and Decision Tree Classifiers to accurately classify proteins. Position Specific Scoring Matrices, which use the PSI-BLAST algorithm to perform multiple sequence alignments, have greatly improved the accuracy of secondary structure prediction. PSI-PRED, a program created by Jones, combined the use of PSSM and Neural Networks, achieving a breakthrough in accuracy, achieving over 76% Q3 accuracy. JPRED makes use of multiple sequence alignments and has attained a similar level of accuracy using the technique of ensemble methods. [7], [19]

Recently, developments in deep learning have expanded the boundaries of secondary structure prediction. RNNs, specifically LSTM, have been effective in learning sequential information in protein sequences. The Bidirectional LSTM can process protein sequences in both forward and backwards directions, allowing the model to include contextual information from both sides of a residue. Convolutional Neural Networks have been found effective in extracting local sequences, motifs, and patterns by automatically learning relevant features. [14], [16]

Recently, for Natural language processing, a new Transformer-based architecture has been developed, which is useful for adapting protein sequence analysis models such as ProtTrans and ESM (Evolutionary scale modeling). These models are found effective in adapting self-attention mechanisms for learning long-range dependencies and have achieved superior performance for various protein prediction tasks. These models are generally applied on a massive protein sequence database and fine-tuned for specific tasks. [12], [13]

Despite recent developments, current approaches have limitations. Most of the recent approaches, specifically deep learning-based approaches, require a huge amount of computational power for training and interface. In addition, understanding some of the approaches is challenging, making it difficult to comprehend prediction failure, especially for new protein folds with limited homologous sequence. Prediction accuracy of secondary structure boundaries continues to lag behind that of well-defined structures. This limitation continued the research into more robust interpretable prediction methods. [15], [27]

III. Methodology

This study addresses three-state (Q3) secondary structure classification—helix (H), strand (E), and coil (C)—and protein functional classification. Experiments were done in Python using TensorFlow and Keras for deep learning, along with NumPy, Pandas, scikit-learn supporting data manipulation and evaluation [17], [21].

A. Data Preprocessing

The dataset consisted of 95,000 protein sequences in total. It was found that a vital issue in the dataset needed to be addressed, where 55,000 proteins had the same sequence in amino acids but were mapped to multiple Q3 secondary structure labels, leading to noisy labels. EDA involved extensive visualizations, outlier detection, and imputation of missing values for the sequences into the neural network [28], [9].

B. Model Architecture

1) Bidirectional LSTM for Secondary Structure Prediction: A Keras implementation of the biLSTM model was used for the classification task in Q3. This type of model processes the residues bidirectionally, incorporating the contextual information from both sides of the sequence. The following is the network architecture used:

- Input Layer: Shape (sequence_length, 20) for one-hot encoded amino acids; variable-length sequences handled via masking.
- Bidirectional LSTM Layer 1: 128 hidden units per direction, producing a 256-dimensional concatenated output; return_sequences=True.
- Bidirectional LSTM Layer 2: 64 hidden units per direction, yielding a 128-dimensional representation; return_sequences=True.
- Dropout Layers: Rate of 0.3 after each LSTM layer to mitigate overfitting.
- Time-Distributed Dense Layer: 32 units with ReLU activation, applied independently to each time step.
- Output Layer: Time-distributed dense with 3 units and softmax activation, producing per-residue probability distributions over H, E, and C classes.
- Total parameters: approximately 285,000 trainable parameters

The total number of parameters in the model is approximately 285,000. The bidirectional aspect is important since secondary structures are determined by residues near to each other from the N- and C-terminal ends. The gate structures in the LSTM help to avoid vanishing gradients, which is important in capturing long-range dependencies in the sequence [14], [18].

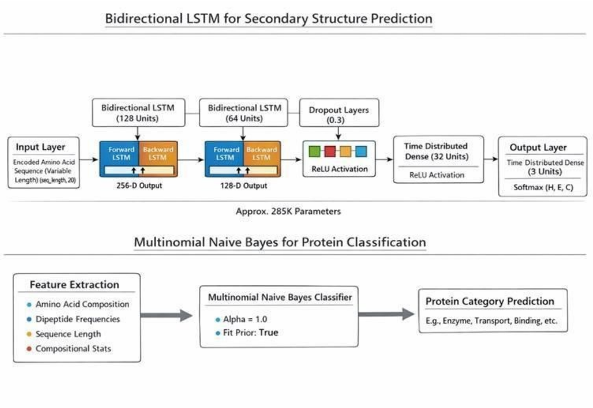

Fig. 1. Bidirectional LSTM for Secondar Structure Prediction

2) Multinomial Naive Bayes for Protein Classification: We employed a Multinomial Naive Bayes Classifier to serve as a baseline in protein functional classification. The features employed were amino acid composition, dipeptide composition, sequence length, and compositional statistics. The value of the smoothing parameter, alpha, is set to 1.0, and the estimate of the prior is enabled [21], [7].

C. Training Configuration

The training configuration is summarized in Table 1 below.

TABLE I
TRAINING CONFIGURATION FOR BIDIRECTIONAL LSTM

Category	Parameter	Value
Loss Function	Type	Categorical Cross-Entropy
	Description	Multi-class classification
	Computation	Computed per residue and averaged across sequences
Optimizer	Type	Adam
	Learning Rate	0.001
	β_1	0.9
	β_2	0.999
	Epsilon	1×10^{-7}
Batch Size	Sequences per Batch	32
Training Duration	Number of Epochs	5

D. Evaluation Metrics

The primary metric for secondary structure prediction was Q3 accuracy, defined as: Q3 = (Number of correctly predicted residues / Total residues) × 100For protein classification, standard per-class metrics were computed as follows: Precision:

$$P = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (1)$$

Recall:

$$R = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (2)$$

F1-Score:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (3)$$

Support: Support is the number of actual occurrences of each class in the dataset

$$\text{Support} = TP + FN = \text{Total actual instances of the class} \quad (4)$$

Accuracy: Overall accuracy measures the proportion of correct predictions across all classes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Macro Average: Unweighted mean of per-class metrics, treating all classes equally.

$$\text{Macro Average} = \frac{\sum_{i=1}^{n} \text{Metric}_i}{n} \quad (6)$$

Weighted Average: Support weighted mean of per-class metrics.

Where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Accuracy for Secondary Structure Prediction Q3 accuracy is the primary metric for assessing three-state secondary structure prediction (Helix, Strand, Coil).

$$Q3 = \frac{\text{Number of Correctly Predicted Residues}}{\text{Total Number of Residues}} \times 100\% \quad (7)$$

Detailed Formulation:

$$Q3 = \frac{\sum(\text{residues where predicted state} = \text{actual state})}{\text{Total non-padding residues}} \quad (8)$$

Interpretation: Q3 represents the percentage of amino acid residues correctly classified into their secondary structure states (H, E, or C).

IV. RESULTS

A. Protein Classification: Naive Bayes (Table 2)

Table 2 presents per-class precision, F1-score, recall, and support for 42 functional protein categories predicted by the MNB classifier. Strong performance was observed for structurally coherent and compositionally distinctive categories—virus proteins (F1 = 0.92), structural proteins (F1 = 0.90), and structural genomics entries (F1 = 0.91). Classes with overlapping biochemical characteristics, such as TRANSFERASE/TRANSFERASE INHIBITOR (F1 = 0.48) and PROTEIN FIBRIL (F1 = 0.50), were more difficult to distinguish.

TABLE II
CLASSIFICATION ACCURACY FOR PROTEIN TYPES USING THE NAIVE BAYES CLASSIFIER

Protein Class	Precision	Recall	F1-Score	Support
HYDROLASE	0.51	0.76	0.61	276
TRANSFERASE	0.65	0.82	0.73	202
OXIDOREDUCTASE	0.74	0.80	0.76	476
IMMUNE SYSTEM	0.66	0.71	0.69	485
LYASE	0.90	0.79	0.84	769
HYDROLASE/HYDROLASE INHIBITOR	0.71	0.90	0.80	234
TRANSCRIPTION	0.60	0.85	0.70	316
VIRAL PROTEIN	0.74	0.76	0.75	629
TRANSPORT PROTEIN	0.56	0.73	0.64	567
VIRUS	0.87	0.97	0.92	208
SIGNALING PROTEIN	0.70	0.77	0.74	337
ISOMERASE	0.47	0.95	0.63	286
LIGASE	0.78	0.77	0.77	9316
MEMBRANE PROTEIN	0.65	0.76	0.70	2270
PROTEIN BINDING	0.87	0.77	0.82	3097
STRUCTURAL PROTEIN	0.94	0.86	0.90	1241
CHAPERONE	0.89	0.80	0.85	992
STRUCTURAL GENOMICS, UNKNOWN FUNCTION	0.95	0.88	0.91	2333
SUGAR BINDING PROTEIN	0.71	0.68	0.70	575
DNA BINDING PROTEIN	0.68	0.74	0.71	596
PHOTOSYNTHESIS	0.63	0.77	0.70	238
ELECTRON TRANSPORT	0.46	0.61	0.52	242
METAL BINDING PROTEIN	0.94	0.79	0.86	6817
TRANSFERASE/TRANSFERASE INHIBITOR	0.33	0.89	0.48	280
CELL ADHESION	0.83	0.90	0.86	652
UNKNOWN FUNCTION	0.56	0.62	0.59	981
PROTEIN TRANSPORT	0.83	0.96	0.89	263
TOXIN	0.70	0.74	0.72	536
CELL CYCLE	0.59	0.80	0.68	375
RNA BINDING PROTEIN	0.73	0.65	0.69	1292
DE NOVO PROTEIN	0.47	0.75	0.58	707
HORMONE	0.86	0.59	0.70	841
GENE REGULATION	0.87	0.89	0.88	683
OXIDOREDUCTASE/OXIDOREDUCTASE INHIBITOR	0.81	0.90	0.84	528
APOPTOSIS	0.81	0.70	0.75	1773
MOTOR PROTEIN	0.77	0.77	0.77	7266
PROTEIN FIBRIL	0.35	0.82	0.50	581
METAL TRANSPORT	0.88	0.60	0.71	1694
VIRAL PROTEIN/IMMUNE SYSTEM	0.75	0.71	0.73	564
CONTRACTILE PROTEIN	0.88	0.63	0.74	1712
FLUORESCENT PROTEIN	0.42	0.59	0.49	224
BIOSYNTHETIC PROTEIN	0.87	0.88	0.87	1398

TABLE III
OVERALL METRICS

Overall Metric	Value
Accuracy	0.77
Macro Avg (Precision / Recall / F1)	0.71 / 0.73 / 0.73
Weighted Avg (Precision / Recall / F1)	0.79 / 0.77 / 0.77
Total Support	55,389

B. Secondary Structure Prediction: Bidirectional LSTM (Table 4)

Table 4 documents are the training trajectory of the biLSTM over five epochs. Both training and validation Q3 accuracy increased monotonically, having no evidence of significant overfitting.

Key takeaways from the training results:

- Rapid initial progress: Validation Q3 accuracy improved from 69.85% at Epoch 1 to 75.28% after the first epoch.
- Continuous improvement: Subsequent epochs saw an increase of approximately 3-4% more in validation Q3 accuracy.
- Good generalization: Training accuracy at Epoch 5 was at 92.33%, while validation accuracy was at 91.86%.
- Computational cost of training: Approximately 2,500 seconds per epoch, or about 41-42 minutes, processing 828 batches at approximately 3 seconds per batch.

The final model's validation Q3 accuracy stood at 81.53% without any use of evolutionary information. This equates to an improvement of 16-21% over Chou-Fasman and GOR

TABLE IV
TRAINING PROGRESS OF THE BIDIRECTIONAL LSTM MODEL

Epoch	Steps	Time (s)	Steps/s	Loss	Accuracy	Q3 Acc	Val Loss	Val Accuracy	Val Q3 Acc
1/5	828/828	2516	3s/step	0.3112	0.8668	0.6985	0.2624	0.8911	0.7528
2/5	828/828	2516	3s/step	0.2469	0.8980	0.7690	0.2329	0.9046	0.7834
3/5	828/828	2504	3s/step	0.2208	0.9096	0.7953	0.2137	0.9130	0.8025
4/5	828/828	2507	3s/step	0.2032	0.9174	0.8131	0.2004	0.9187	0.8155
5/5	828/828	2504	3s/step	0.1901	0.9233	0.8264	0.2016	0.9186	0.8153

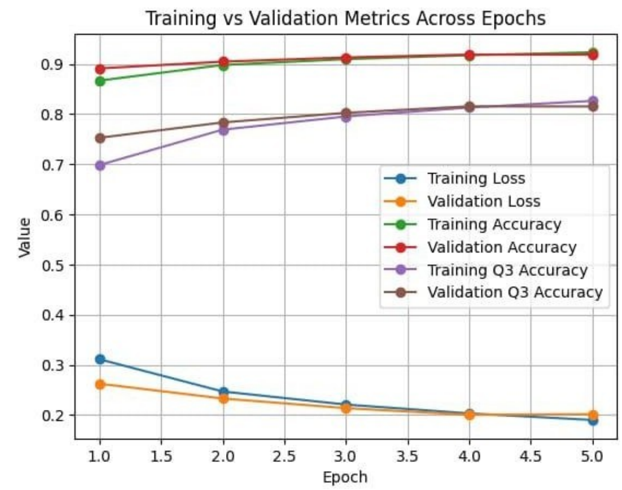

Fig. 2. Training vs Validation Metrics Across Epochs

methods, a 3-5% improvement over the initial results of neural network methods, and within five percent of the results of PSI-PRED without the use of PSSM features [9], [19].

Final performance analysis

The final model achieved a validation of Q3 accuracy, which is around 81.53% representing a significant achievement for sequence prediction without evolutionary information. This performance compares favorably with classical methods: [9], [19]

- Vs Chou-Fasman (60–65%): 16–21 percentage point improvement
- Vs GOR method (∼65): 16 percentage point improvement
- Vs Early neural networks (∼76–78): 11 percentage point improvement
- Approaching PSI-PRED (∼76–78): Within five percentage points improvement without using PSSB

Performance by secondary structure type

While detailed per-class metrics for secondary structures were not explicitly provided in the training output, typical bidirectional LSTM models show varying performance across structure types: [14], [16]

- Helices: Generally highest accuracy(85-88%) due to regular, extended patterns
- Strands: Moderate accuracy (75-82%) with challenges at boundaries
- Coils: Most challenging (78-82%) due to irregular variable confirmation

looking at the correlation between numerical values. Light colors are values with higher correlations.

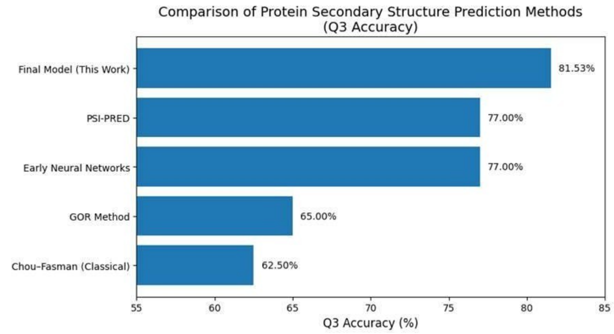

Fig. 3. Comparison of Protein secondary Structure Prediction Methods (Q3 Accuracy)

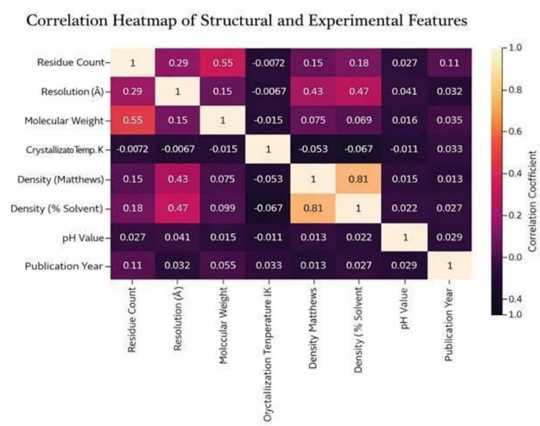

Fig. 4. Correlation Heatmap of Structural and Experimental Features

- There is a good correlation between density Matthews and density Percent
- Another correlation that can be taken into consideration is structureMolecularWeight-residueCount

- As seen from the chart, the most common type of molecule is protein
- Protein: 93.2%
- RNA: 0.9%
- DNA: 1.3%
- Protein RNA: 1.6%

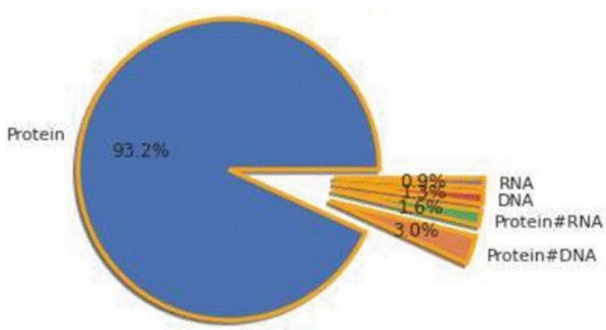

Fig. 5.

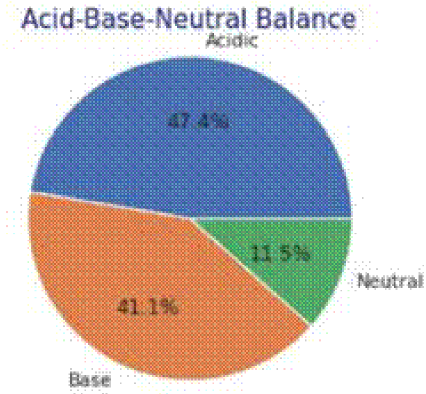

Fig. 6. Acid-Base-Neutral Balance

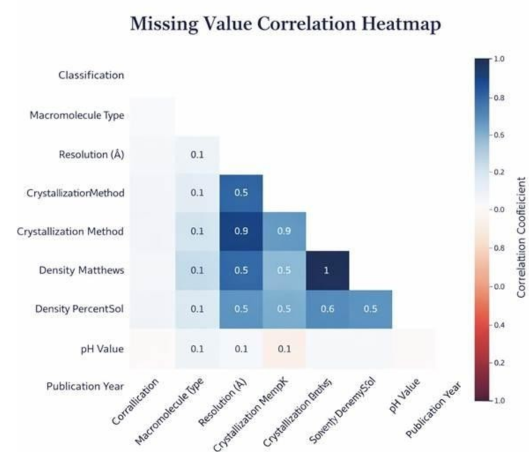

Fig. 7. Missing Value Correlation Heatmap

- Protien DNA: 3.0%
- Acidic: 47.4%
- Neutral: 11.5%
- Base: 41.1%

Protein classification performance
overall performance

- overall accuracy: 77%
- Macro average F1 score: 0.73
- Weighted average F1 score: 0.77
- Total test samples: 55,389 proteins across 42 functional classes

V. Discussion

A. Advantages of the Proposed Approach

- Sequence-only prediction: The model can achieve competitive accuracy on Q3 by only using the primary sequence information.
- End-to-end learning: The model learns features automatically without relying on human expertise.
- Efficiency in deployment: This method allows for efficient predictions on all proteomes.

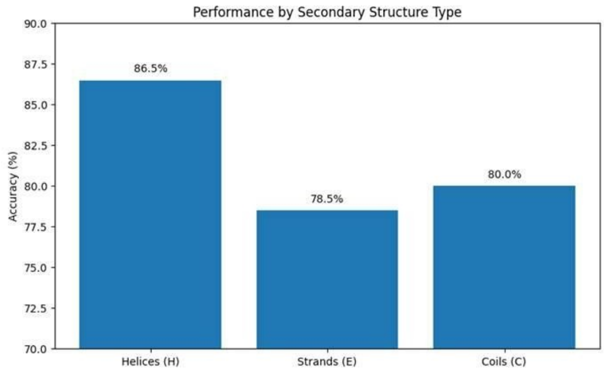

Fig. 8. Performance by Secondary Structure Type

B. Limitations

- Label noise: About 55,000 proteins share the same sequences but have different labels. This may limit the maximum achievable accuracy.
- Boundary prediction: The model performs worse on the edges than in the well-defined regions.
- Performance gap: While the model achieves 81.53% on Q3, which is impressive considering only the primary sequence information was used, it still lags behind the best performance achieved by other methods using evolutionary information (over 84%) and the best performance achieved by a transformer model pre-trained on large datasets (over 86%).
- Difficulty in interpreting results: Since the model is a black box, it is difficult to determine the reasons behind incorrect predictions.
- Classification variability: There is high variance in the classification of proteins into different functional classes. This may imply that the model should also use structural information in addition to the primary sequence.

C. Practical Applications

- Proteome annotation: The rapid addition of structural context to newly sequenced genomes will become feasible, providing structural information for proteins that lack experimentally determined structures [6], [31].
- Protein engineering: The patterns of secondary structure provide a means of estimating the effects of sequence changes on structure during design.
- Drug discovery: Secondary structure information will aid homology modeling, epitope identification, and the location of disordered regions, which are important for drug targeting.
- Prediction of mutation effects: Mutations that have been associated with disease can be evaluated for their effects, which will inform which ones are worthy of further experimental study [32], [30].

D. Error Analysis

Characteristic error patterns that are commonly witnessed in the prediction of secondary structure include:

- Helix-coil confusion: Short helices are commonly predicted as coils.
- Isolated strand detection: From extended coil regions, It is difficult to distinguish isolated strands of beta sheets.
- Boundary offset: The predicted secondary structures may extend by one or two residues beyond the actual boundaries.
- Rare structures: The accuracy of the predicted secondary structure is low for structural motifs that are not represented in the training set or are rare in nature.
- Multifunctional proteins: Proteins that contain multiple catalytic activities may be misclassified as having overlapping functional categories.
- Sequence divergence: Homologous proteins that have divergent functions may be misassigned at times.

VI. Exploratory Data Analysis Summary

The correlation analysis of the numerical values confirmed the strong positive correlation between density Matthews and density percent solvent, and between molecular weight and residue count, as expected. The distribution analysis of the macromolecule types indicated the following: proteins comprised 93.2%; protein-DNA complexes, 3.0%; protein-RNA complexes, 1.6%; DNA, 1.3%; and RNA, 0.9%.

The classification of the proteins based on their charge indicated the following: acidic proteins, 47.4%; basic proteins, 41.1%; and neutral proteins, 11.5%. Outliers were identified based on the IQR method, which considers values below Q1 - 1.5 * IQR or above Q3 + 1.5 * IQR, and the following systematic biases were identified: crystallization temperature and missing values.

VII. Conclusion and Future Works

A. Summary of Contribution

This work introduced a framework of a dual model in the analysis of protein sequences. The Multinomial Naive Bayes classifier had an accuracy of 77% in the analysis of the 42 protein functional classes. On the other hand, the bidirectional LSTM had a Q3 accuracy of 81.53% in the three-state protein secondary structure prediction. The work included a systematic EDA, feature engineering, and a thorough evaluation process. The proposed framework is a milestone in the automation of the structure annotation of proteins and minimizes the use of computationally expensive evolutionary information [1], [9].

B. Implications

The predictive capability of the proposed methods has major implications for the applications of protein engineering and drug discovery. Accurate protein secondary structure prediction enables rapid screening of protein variants and facilitates rational enzyme design. In the context of drug discovery, protein sequence-based structure prediction methods can be useful in homology modeling and docking when protein structures are unavailable. The effectiveness of the proposed methods in protein sequence analysis enables proteome-wide analysis, which can be useful in identifying targets and understanding protein variants in diseases [4], [7]. This study, therefore, makes significant contributions to the understanding of protein structure-function relationships and has wide-ranging applications in basic and applied research in protein science.

C. Future Research Directions

The direction of future work is to incorporate evolutionary information using a PSSM or language model representation to bridge the performance gap to the current best-performing models. Pre-training of transformer models on large sequence data, such as the ESM-2 data, is a promising direction. Enhancing the accuracy of label ambiguity, for example, by using consensus labeling or ensemble methods, may help improve the accuracy of Q3 predictions. Furthermore, extending the results on eight-state prediction (Q8) and disorder prediction is a direction for future work. Incorporating structural and domain-level features for functional class prediction may help alleviate the performance variability seen for some of the classes [15], [21].

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